

Cacti with Arbitrarily Many Triangles and One Hostile Cycle

28 July 2026

Abstract

Let G be a connected cactus whose cyclic blocks are $r \geq 1$ triangles and one cycle C_q , where $q = 4k + 1 \geq 5$. The cyclic blocks may share cut vertices in arbitrary incidence topology, bridge paths may have arbitrary length, and arbitrary finite trees may be attached throughout. We prove

$$s^+(G) > |V(G)|.$$

The proof is rank-uniform. A maximum-packing Voronoi partition proves strict positivity for every triangular cactus and yields a rooted guard theorem for one hostile cycle. At the global level, actual bridges remove triangular leaf clusters inductively. In the remaining shared-cut cluster, one interface is handled by the rooted theorem, while two or more interfaces allow the hostile cycle itself to be split into consecutive intervals, leaving only strict triangular induced territories. Every cut, connector remnant, and attached tree receives exactly one owner.

1 Statement and basic facts

All graphs are finite, simple, and undirected. For the adjacency eigenvalues $\lambda_1, \dots, \lambda_n$ of a graph H , put

$$s^+(H) = \sum_{\lambda_i > 0} \lambda_i^2, \quad s^-(H) = \sum_{\lambda_i < 0} \lambda_i^2, \quad \sigma(H) = s^+(H) - |V(H)|.$$

A cactus is a graph whose blocks are bridges, isolated vertices, or cycles. Its cyclomatic number is the number of cyclic blocks.

Theorem 1.1. *Let G be a connected cactus whose cyclic blocks are $r \geq 1$ copies of C_3 and one copy of $Q = C_q$, where $q = 4k + 1 \geq 5$. Allow arbitrary bridge connectors and arbitrary finite trees attached at arbitrary vertices. Then*

$$\boxed{s^+(G) > |V(G)|}.$$

We use square-energy superadditivity on induced vertex partitions.

Lemma 1.2 (Induced-partition superadditivity). *If $V(H) = V_1 \dot{\cup} \dots \dot{\cup} V_t$, then*

$$s^+(H) \geq \sum_{j=1}^t s^+(H[V_j]), \quad \sigma(H) \geq \sum_{j=1}^t \sigma(H[V_j]).$$

Proof. For a real symmetric matrix A , its positive spectral part satisfies

$$\mathrm{tr}(A_+^2) = \max_{X \succeq 0} (2 \mathrm{tr}(AX) - \mathrm{tr}(X^2)).$$

This follows by diagonalizing A : deleting the off-diagonal entries of the positive semidefinite test matrix preserves positive semidefiniteness and only increases the objective, after which the scalar maximizers are $x_i = \max(\lambda_i, 0)$.

Order the vertices by the partition, let $A = A(H)$, and put $B = \text{diag}(A(H[V_1]), \dots, A(H[V_t]))$. Testing the variational formula with $X = B_+$ gives

$$s^+(H) \geq 2 \text{tr}(AB_+) - \text{tr}(B_+^2) = \text{tr}(B_+^2) = \sum_j s^+(H[V_j]),$$

because B_+ is block diagonal and the off-block entries of A do not contribute to the trace. Subtracting the partitioned vertex count proves the second inequality. $\square$

2 Triangular territories

For a graph F of order n , put

$$\Psi_F(t) = i^{-n} \det(itI - A(F)) = \prod_{j=1}^n (t + i\lambda_j), \quad t > 0,$$

and define its continuous phase, tending to zero at infinity, by

$$\Theta_F(t) = \sum_{j=1}^n \arctan \frac{\lambda_j}{t}.$$

Lemma 2.1 (Signed Coulson identity). *For every graph F ,*

$$s^+(F) - s^-(F) = -\frac{4}{\pi} \int_0^\infty t \Theta_F(t) dt.$$

Proof. For real λ ,

$$\lambda|\lambda| = \frac{2}{\pi} \int_0^\infty \frac{\lambda^3}{\lambda^2 + t^2} dt.$$

Also $\Theta'_F(t) = -\sum_j \lambda_j / (\lambda_j^2 + t^2)$. Since $\sum_j \lambda_j = \text{tr } A(F) = 0$,

$$\sum_j \frac{\lambda_j^3}{\lambda_j^2 + t^2} = t^2 \Theta'_F(t).$$

Sum the scalar identity and integrate by parts. The boundary term vanishes: $t^2 \Theta_F(t) \rightarrow 0$ at zero, while the arctangent expansion gives $\Theta_F(t) = O(t^{-3})$ at infinity. These estimates also give convergence. $\square$

The cycle-packing number $\nu(H)$ is the maximum number of pairwise vertex-disjoint cycles in H .

Lemma 2.2 (Packing-one triangular packet). *Let H be a connected cactus whose $b \geq 1$ cyclic blocks are triangles and whose cycle-packing number is one. Arbitrary trees may be attached. Then*

$$s^+(H) > s^-(H), \quad \sigma(H) > b - 1 \geq 0.$$

Proof. For $t > 0$, let

$$Z_F(t) = \sum_M t^{|V(F)|-2|M|}, \quad \Psi_H(t) = i^{-|V(H)|} \det(itI - A(H)),$$

where M ranges over matchings. Grouping the Sachs expansion by collections of vertex-disjoint cycles, a triangle has multiplier $-2i$. Because no two triangles are disjoint,

$$\Psi_H(t) = Z_H(t) - 2i \sum_T Z_{H-V(T)}(t).$$

Every displayed matching partition is positive. Hence the continuous argument $\Theta_H(t)$ of $\Psi_H(t)$, tending to zero at infinity, lies strictly in $(-\pi/2, 0)$. Lemma 2.1

$$s^+(H) - s^-(H) = -\frac{4}{\pi} \int_0^\infty t \Theta_H(t) dt$$

therefore gives strict positivity. Since $|E(H)| = |V(H)| - 1 + b$ and $s^+(H) + s^-(H) = 2|E(H)|$, the surplus exceeds $b - 1$. $\square$

Lemma 2.3 (Maximum-packing territories). *Let A be a connected cactus all of whose cyclic blocks are triangles. Let $T_0, \dots, T_{m-1}$ be a maximum-cardinality collection of pairwise vertex-disjoint triangles. Assign each vertex v to the lexicographically least pair*

$$(d_A(v, V(T_j)), j),$$

and let A_j be the induced graph on the vertices assigned to j . Then each A_j is connected, contains T_j , and has cycle-packing number one. If a root z is prescribed and T_0 is chosen nearest z , then $z \in A_0$.

Proof. A predecessor of an assigned vertex on a shortest path to its selected cycle has the same owner: distances are one-Lipschitz, and the priority rule resolves ties consistently. Thus every territory is connected and contains its selected cycle. If A_j contained two disjoint cycles D, E , then

$$D, E, \{T_h : h \neq j\}$$

would be a packing larger by one. Notice that D, E replace T_j ; they need not also be disjoint from it. The root assertion follows from choosing T_0 with minimum root distance and giving it first priority. $\square$

Corollary 2.4 (All triangular cacti). *Every connected cactus containing at least one cyclic block, all cyclic blocks being triangles, satisfies $\sigma(A) > 0$, with arbitrary attached trees.*

Proof. Apply Lemma 2.3. Every territory is a packing-one triangular cactus retaining at least one triangle, so Lemma 2.2 gives strictly positive surplus to every territory. Sum by Lemma 1.2. $\square$

3 One rooted hostile cycle

Put

$$\delta_q = \sec \frac{\pi}{q} - 1.$$

For $q \geq 5$, $0 < \delta_q < 1$.

Lemma 3.1 (Packing-one hostile packet). *Let H be a connected cactus whose cyclic blocks are $Q = C_q$ and $a \geq 1$ triangles, where $q \equiv 1 \pmod{4}$. Suppose no two triangles are vertex-disjoint. The cycle Q may meet the triangular part at one cut vertex or be joined to a distinguished root of that part by one path. Arbitrary trees may be attached anywhere. Then*

$$s^+(H) - s^-(H) > -2\delta_q, \quad \sigma(H) > a - \delta_q \geq 1 - \delta_q > 0.$$

Proof. Let the *spine* be the subgraph consisting of all cyclic blocks and the unique minimal paths joining them. Every component outside it is a tree with one spine attachment; two attachments would create another cycle. Orient each off-spine branch toward the spine. If $u \rightarrow v$ is an oriented branch edge, let $T_{u \rightarrow v}$ be the component below u , including u , and define

$$M_{u \rightarrow v}(t) = \frac{Z_{T_{u \rightarrow v}}(t)}{Z_{T_{u \rightarrow v} - u}(t)}.$$

Splitting matchings according to whether u is unmatched or matched to one child proves the exact recursion

$$M_{u \rightarrow v}(t) = t + \sum_{w \text{ child of } u} \frac{1}{M_{w \rightarrow u}(t)} \geq t. \quad (1)$$

The leaf case is $M = t$. Extracting all denominators recursively gives a common positive factor $K(t)$ and replaces the unmatched-vertex activity at a spine vertex by

$$\alpha_v(t) = t + \sum_{u \text{ branch child of } v} \frac{1}{M_{u \rightarrow v}(t)} = t + y_v(t), \quad y_v(t) \geq 0. \quad (2)$$

This is an exact matching decomposition.

For positive activities $\alpha = (\alpha_v)$, define

$$Z_F(\alpha) = \sum_{M \text{ matching of } F} \prod_{v \text{ unmatched by } M} \alpha_v.$$

It is positive and coordinatewise nondecreasing. Tree elimination says that each Sachs matching carrier is $K(t)$ times the corresponding weighted spine carrier. Let $S = V(Q)$. The grouped Sachs expansion has the form

$$\frac{\Psi_H(t)}{K(t)} = R + 2i(B - A),$$

where

$$B = Z_{\text{spine}-S}(\alpha) > 0, \quad A = \sum_T Z_{\text{spine}-V(T)}(\alpha) > 0,$$

and

$$R = Z_{\text{spine}}(\alpha) + 4 \sum_{T \cap Q = \emptyset} Z_{\text{spine}-(S \cup V(T))}(\alpha) > 0.$$

The normalized Sachs multiplier of C_ℓ is $-2i^{-\ell}$. Thus a triangle contributes $-2i$, while $q \equiv 1 \pmod{4}$ makes Q contribute $+2i$. A Sachs collection uses vertex-disjoint cycles, so it contains at most one triangle. The empty collection gives $Z_{\text{spine}}(\alpha)$, singleton Q and triangle collections give $2iB$ and $-2iA$, and a disjoint Q -triangle pair gives $(2i)(-2i) = 4$. These exhaust the expansion and prove the displayed formulas.

Let $Z_q(t)$ denote the unweighted matching partition of C_q . Partition spine matchings according to whether they use an edge between S and its complement. Those using no such edge factor into a

matching on Q and one on the vertex-deleted remainder; all remaining matchings have nonnegative weight. Thus

$$Z_{\text{spine}}(\alpha) = Z_Q(\alpha|S)B + E, \quad E \geq 0.$$

Coefficient positivity and $\alpha_v \geq t$ give $Z_Q(\alpha|S) = Z_q(t) + L$ with $L \geq 0$. Therefore

$$R - Z_q(t)(B - A) = E + LB + Z_q(t)A + 4 \sum_{T \cap Q = \emptyset} Z_{\text{spine}-(S \cup V(T))}(\alpha) > 0. \quad (1)$$

Since $K(t) > 0$ it changes no phase. Since $R > 0$, the continuous phase is the principal value

$$\Theta_H(t) = \arctan \frac{2(B - A)}{R}.$$

The isolated hostile cycle has $\Psi_Q(t) = Z_q(t) + 2i$ and phase $\theta_q(t) = \arctan(2/Z_q(t))$. If $B - A \leq 0$, then $\Theta_H \leq 0 < \theta_q$; if $B - A > 0$, inequality (1) gives the same strict comparison. Thus $\Theta_H(t) < \theta_q(t)$ for all $t > 0$.

Lemma 2.1 and the exact cycle value

$$s^+(C_q) - s^-(C_q) = -2 \left(\sec \frac{\pi}{q} - 1 \right)$$

give the first displayed inequality. Finally $|E(H)| = |V(H)| + a$ and $s^+(H) + s^-(H) = 2|E(H)|$, proving the surplus bound. $\square$

Lemma 3.2 (Exact hostile-cycle value). *For $q = 4k + 1 \geq 5$,*

$$s^+(C_q) - s^-(C_q) = -2 \left(\sec \frac{\pi}{q} - 1 \right).$$

Proof. The eigenvalues are $2 \cos(2\pi j/q)$, $0 \leq j < q$. The positive ones have $j = 0, 1, \dots, k$ and $j = q - k, \dots, q - 1$, so

$$s^+(C_q) = 4 \left(1 + 2 \sum_{j=1}^k \cos^2 \frac{2\pi j}{q} \right).$$

Using $\cos^2 x = (1 + \cos 2x)/2$ and

$$\sum_{j=1}^k \cos(jx) = \frac{\sin(kx/2) \cos((k+1)x/2)}{\sin(x/2)}$$

at $x = 4\pi/q$ gives $s^+(C_q) = q + 1 - \sec(\pi/q)$. Since $s^+(C_q) + s^-(C_q) = 2q$, the claimed difference follows. $\square$

Theorem 3.3 (Rooted all-rank guard). *Let A be an arbitrary connected triangular cactus with at least one triangle, rooted at z . Attach $Q = C_q$, $q \equiv 1 \pmod{4}$, to z through exactly one interface: identify a vertex of Q with z , or join them by one path. Allow arbitrary attached trees. Then*

$$\sigma(G) > 1 - \delta_q > 0.$$

Proof. Apply Lemma 2.3 to a maximum triangle packing of A , choosing the root territory first. Add Q , the whole connector, and all trees rooted there to the root territory. Its retained triangles have packing one and include the selected root triangle, so Lemma 3.1 gives surplus $> 1 - \delta_q$. Every other induced territory is a packing-one triangular cactus and has positive surplus by Lemma 2.2. The unique-interface hypothesis ensures that no connector vertex must have two owners. Sum by Lemma 1.2. $\square$

4 Shared-cut clusters and ownership

Two cyclic blocks are adjacent when they share a cut vertex. A *shared-cut cluster* is a maximal connected component under this adjacency. In the cactus block–cut tree, the *support* of a cluster is the connected subtree consisting of its cyclic-block nodes and every cut-vertex node joining adjacent cyclic blocks in the cluster. Distinct supports are disjoint, since a cut belonging to both would join the clusters. Contract each full support to one cluster node. In the resulting tree take the minimal subtree spanning all cluster nodes and suppress each noncluster vertex of degree two. This is the *reduced cluster tree*. Every leaf is a cluster node, and every reduced edge contains a nonempty chain of actual bridge blocks; otherwise its endpoint cycles would lie in one cluster.

When one actual bridge in such a chain is cut, its two endpoint components give a vertex partition into connected induced cacti. Every remaining connector segment and every off-hull tree follows the side containing its unique attachment. An off-hull component cannot have two hull attachments: together with the hull path they would create another cycle. This is the ownership rule used below.

For one shared-cut cluster, form its bipartite cycle–cut incidence graph I : cycle nodes are cyclic blocks, cut nodes are vertices belonging to at least two cyclic blocks, and containment gives incidence. This graph is a tree. Indeed, an incidence cycle would realize either a cycle using several blocks or two blocks meeting in more than one vertex, both impossible in a cactus.

Lemma 4.1 (Splitting the distinguished cycle). *Suppose one shared-cut cluster consists of Q and triangles. If Q is incident in I with $d \geq 2$ distinct cut nodes, then the graph has an induced vertex partition into d connected nonempty triangular cacti.*

Proof. Let $x_1, \dots, x_d$ be the distinct occupied vertices of Q in cyclic order. Partition $V(Q)$ into d nonempty proper consecutive intervals J_i , each containing x_i and no other occupied vertex. Consecutive occupied vertices may receive singleton intervals; the construction also works when all vertices of Q are occupied.

Deleting the Q node from I gives exactly d components K_i , with x_i in K_i . The *realization* of K_i consists of all private vertices of its cycle blocks and all its cut vertices. Give one territory the interval J_i , the complete realization of K_i , and every tree whose unique attachment it owns. Since x_i is a shared cyclic cut, K_i contains at least one triangle. Each territory is connected through x_i . The intervals partition Q , the components K_i partition the remaining incidence tree, and all multiway branches at one cut remain with one owner. Thus the territories are induced, disjoint, and exhaustive. Every J_i is a proper path, so Q is retained nowhere; all retained cycles are triangles. $\square$

5 Proof of the main theorem

Proof of Theorem 1.1. Induct on the number N of shared-cut clusters.

Suppose first that $N > 1$. Only one cluster contains Q . The reduced cluster tree has at least two leaves, so it has a leaf cluster L not containing Q . Cut the first actual bridge away from L , using the ownership rule above. One induced side A is a connected triangular cactus, hence $\sigma(A) > 0$ by Corollary 2.4. The other side B contains Q and all remaining clusters.

If B retains at least one triangle, it is a smaller instance of the theorem, so induction and Lemma 1.2 give strict positivity. If B retains no triangle, there is no other triangular cluster, so $N = 2$, L contains all triangles, and the original graph consists of a connected triangular cactus joined to Q through exactly one bridge interface. Theorem 3.3 closes this endpoint directly. This avoids the invalid step of asking qualitative triangular strictness to pay the negative surplus of an isolated hostile cycle.

It remains to treat $N = 1$. Let d be the degree of the Q node in the cycle–cut incidence tree. Since at least one triangle is present and the cluster is connected, $d \neq 0$.

If $d = 1$, deleting the Q node leaves one connected triangular cactus rooted at the unique shared cut. The cycles need not all contain that root, so a common-cut bouquet theorem would be insufficient; Theorem 3.3 applies exactly and gives strict positivity.

If $d \geq 2$, Lemma 4.1 partitions G into connected nonempty triangular cacti. Every part has positive surplus by Corollary 2.4, and Lemma 1.2 completes the proof. $\square$

6 Scope

The proof is uniform in the number of triangular blocks and does not enumerate incidence trees. It depends essentially on there being one distinguished hostile cycle. With two pentagons, a maximum-packing partition can retain the two hostile cycles while splitting every triangular guard; already the incidence path $C_5 - x - C_3 - y - C_5$ exhibits this ownership obstruction. Thus the present theorem does not claim the arbitrary $T^r C_5 C_5$ family.

References

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- [2] C. A. Coulson, *On the calculation of the energy in unsaturated hydrocarbon molecules*, Proc. Cambridge Philos. Soc. 36 (1940), 201–203.